MGNREGA Project Data Pipeline Documentation

Prepared from: project notebooks and CSVs in this folder.

Final panel file: `Final Datasets/final_panel_data.csv`

Final panel shape: 3,223 rows, 29 columns (293 districts x 11 years)

Final panel years: 2014-2024 (annual frequency)

1) Data Sources and Coverage

Variable Group	Primary Source	Processed File Used in Panel	Raw/Prepared Coverage Before Final Intersection
Income	CMIE Consumer Pyramids (household-level income)	`Final Datasets/district_year_income.csv`	528 districts, 2014-2025 (annual)
Consumption	CMIE Consumer Pyramids (household-level adjusted expenditure)	`Final Datasets/annual_consumption_by_district.csv`	528 districts, 2014-2025 (annual)
Asset Ownership	CMIE Consumer Pyramids	`Final Datasets/district_asset_index.csv`	526 districts, 2014-2025 (annual)

Variable Group	Primary Source	Processed File Used in Panel	Raw/Prepared Coverage Before Final Intersection
	asset indicators		
Crop Yield / Agriculture	Directorate of Economics and Statistics (India), crop-wise district files	`Final Datasets/district_agriculture_panel.csv`	637 districts, 2013-2025 (annual)
MPI	NITI Aayog district MPI reporting (NFHS-based)	`Final Datasets/District-Level MPI.csv`	704 districts (static district-level; no year)
Population demographics	Census 2011 (rural population and area)	`Final Datasets/rural_population_area.csv`	640 districts (static district-level; no year)
Rainfall	IMD Pune gridded rainfall NetCDF + district polygons	`Final Datasets/district_rainfall_2014_2025.csv`	659 districts, 2014-2025 (annual totals in mm)
MGNREGA (Employment / Wage / SC-ST)	MGNREGA web portal scraping (Outlays & Outcomes tables)	`Final Datasets/mgnrega_employment.csv` , `mgnrega_wage.csv` , `mgnrega_sc-st.csv`	Employment: 760 districts; Wage: 764; SC-ST: 754; years 2014-2025

2) Source-Wise Preprocessing and Processing

CMIE Income (District-Year)

Notebook: `Income/Income.ipynb`

- Input loaded from `Income.csv` (household-level CMIE records).
- Parsed month strings into datetime, then extracted `YEAR` .
- Built weighted income variable: `TOT_INC * R_HH_WGT_MS` .
- Aggregated by `DISTRICT, YEAR` .
- Computed weighted district-year mean income as `weighted_tot_inc_sum / weights_sum` .
- Exported to district-year file (`district_year_income.csv` ; later standardized in final folder).

Final panel variable: `Total_Income` (renamed from source in panel merge)

CMIE Consumption (District-Year)

Notebook: `Consumption/Consumption-Aggregate.ipynb`

- Input loaded from `ConsumForCAPM.csv` .
- Used household expenditure fields (especially `ADJ_TOT_EXP`).
- Built annual district aggregation and saved as `annual_consumption_by_district.csv` .
- Notebook also includes monthly and national-level aggregation explorations.

Final panel variable: `ADJ_TOT_EXP`

CMIE Asset Ownership Index

Notebook: `Asset Ownership/Asset Onwership.ipynb`

- Constructed district-year asset ownership percentages using household weights.
- Created binary indicators for available household assets.
- Aggregated weighted asset possession rates at `STATE-DISTRICT-YEAR` .
- Applied PCA on asset percentage variables; first component used as asset index.
- Min-max normalized PCA index to 0-1 scale as `asset_index_0_1` .

- Exported `DISTRICT, YEAR, asset_index_0_1` as `district_asset_index.csv`.

Final panel variable: `asset_index_0_1`

Crop Yield / Agriculture Panel (DES India)

Notebook: `Crop Yeild Data/aggregator.ipynb`

- Loaded crop-wise raw files (rice, cotton, arhar, gram, pulses, sugarcane).
- Converted area/production numeric strings (including comma cleanup).
- Forward-filled state/district labels where blanks existed after multi-row table formats.
- Cleaned numbering prefixes from district/state names.
- Merged all crop tables by `State, District, Year` (outer merge).
- Constructed totals:
  - `total_area = rice + cotton + arhar + sugarcane area`
  - `total_production = rice + cotton + arhar + sugarcane production`
- Filtered districts requiring required year continuity; added future years then linearly interpolated by district.
- Computed `agri_yield_index = total_production / total_area`.
- Exported as `district_agriculture_panel.csv`.

Final panel variables: `total_area`, `total_production`, `agri_yield_index`

Rainfall (IMD Pune)

Notebook: `Rainfall Data/data-extraction.ipynb`

- Loaded district boundaries from `india_districts_merged.geojson`.
- Read annual IMD NetCDF files (`RF25_ind{year}_rfp25.nc`) for 2014-2025.
- Summed daily gridded rainfall into annual raster totals.
- Computed district-level zonal mean annual rainfall (mm) using raster stats.
- Exported district-year rainfall to `district_rainfall_2014_2025.csv`.

Final panel variable: `Annual_Rainfall_mm`

MPI (NITI Aayog) and Census 2011 Population

Integrated in: `Final Datasets/panel_data_construction.ipynb`

- MPI read from `District-Level MPI.csv` .
- MPI columns standardized; duplicate district rows averaged (to combine NFHS-based variants).
- Population read from `rural_population_area.csv` , numeric cleanup for comma-formatted values.
- Population renamed to `Rural_Population` and `Area_sqkm` (where available).
- Both treated as static district-level datasets and left-joined after panel construction.

Final panel variables: `NFHS-4_MPI` , `NFHS-5_MPI` , `Rural_Population` , `Area_sqkm`

3) MGNREGA Scraping and Engineering (Detailed)

Main notebook: `MGNREGA DATA/MGNREGA R7 1.2/MGNREGA-Web-scrapper.ipynb`

Supporting merge notebook: `MGNREGA DATA/MGNREGA R7 1.2/data-merge-1.ipynb`

3.1 Scraping design

- Uses Selenium + BeautifulSoup.
- Automates login/captcha by reading hidden captcha value (`hfCaptcha`) and submitting it.
- Enumerates financial years from dropdown and filters to 2014-15 onward.
- For each year and each state:
    - Re-opens page to avoid stale session state.
    - Selects year and state dropdowns.
    - Clicks `Outlays & Outcomes` report link.
    - Parses resulting HTML tables; picks largest table as district report table.
    - Selects required columns by position index and renames them to wage-related fields.
    - Adds year and state columns, drops table-total rows, and saves CSV per state-year.
- Files are stored in `MGNREGA_DATA/{year}/{STATE}.csv` .

3.2 Robustness and recovery cycle

- A missing-job scanner checks all expected year-state files.
- Writes missing combinations to `missing_jobs.txt`.
- Recovery loop re-runs failed year-state jobs one by one in fresh browser sessions.
- Notebook output indicates targeted recovery for states/years that initially failed table extraction.

3.3 Post-scrape consolidation

- `data-merge-1.ipynb` merges all state CSVs into one file per year in `MGNREGA_DATA/Merged`.
- Additional cleaning removes rows with invalid/non-numeric second column values in merged outputs.
- These merged outputs are the base for MGNREGA panel indicators used later in final datasets.

3.4 SC-ST interpolation patch

- In `Final Datasets/panel_data_construction.ipynb`, missing SC-ST values were explicitly patched for Karnataka, West Bengal, and Ladakh.
- West Bengal district aliases were consolidated first (old/new naming harmonization).
- District-year full grids (2014-2024) were created for target states and missing values linearly interpolated.
- Negative values clipped to 0 and rounded.

4) How Final Panel Data Was Built

Notebook: `Final Datasets/panel_data_construction.ipynb`

1. **Load and audit** all panel datasets (MGNREGA, agriculture, asset, rainfall, income, consumption).
2. **Normalize keys:** trim/uppercase district names; parse years from financial-year strings.
3. **Apply district alias map** for cross-source harmonization (state split names, old/new district spellings, etc.).
4. **Fuzzy diagnostics** run to identify unresolved district naming mismatches.
5. **Define common years:** target years 2014-2024, then intersect available years across all panel datasets.
6. **Balanced district selection:** keep only districts that have all common years in every panel dataset.
7. **Sequential inner merge** on `District, Year` for:

   - Employment -> Wage -> SC-ST -> Agriculture -> Asset -> Rainfall -> Income -> Consumption
8. **Attach static data** (MPI + Census population/area) using district-level left joins.
9. **Quality checks:** duplicate district-year checks, balance checks, missing-value diagnostics, coverage plots.
10. **Export outputs:** `final_panel_data.csv` and `common_districts.csv`.

5) Final Panel: Time Period, Frequency, and Variables

Time period: 2014 to 2024

Frequency: Annual district panel

Panel extent: 293 districts, 21 states, balanced over 11 years

Row count check: 293 x 11 = 3,223 rows (balanced panel)

Variables included:

- MGNREGA labor indicators (households, employment demand/availing, persondays, women participation, SC/ST workers)
- MGNREGA wage and expenditure indicators
- Agriculture totals and yield index
- Asset ownership index (0-1)
- Annual rainfall
- Income and consumption (CMIE-derived district-year aggregates)
- Static MPI and Census rural demographics

6) Notes and Interpretation Guidance

- The final panel is a strict intersection panel, so district count is lower than source-specific coverage.
- Static variables (MPI, Census population/area) do not vary by year but are repeated across years after merge.
- Some components required interpolation/harmonization (notably SC-ST MGNREGA series and agriculture future-year completion).

- District naming harmonization is a major contributor to successful merging; alias maps are core to reproducibility.